

Chapter 3: Hedging Strategies Using Futures (Part 3)

4. Cross Hedging and Optimal Hedge Ratio (Section 3.4)

A. What is Cross Hedging?

- **Definition:** Cross hedging occurs when the asset underlying the futures contract is **different** from the asset whose price is being hedged.
- **Necessity:** This is necessary when there is no actively traded futures contract that exactly matches the exposed asset.
- **Example:** An **airline** needs to hedge the price of **jet fuel**. Since jet fuel futures are not actively traded, the airline might use **heating oil futures contracts** because their prices are closely related.

B. Why a Hedge Ratio $\neq 1.0$ is Necessary

- **Hedge Ratio:** The ratio of the size of the position taken in futures contracts ($Q_F \times N$) to the size of the real-world exposure (Q_A).
- **Problem:** In cross hedging, a 1.0 ratio (e.g., hedging 100 barrels of jet fuel with 100 barrels of heating oil) is usually **suboptimal**. Since the prices don't move perfectly together, you need to adjust the amount of the hedging asset to match the exposure.
- **Goal:** The hedger must choose the value for the hedge ratio (h) that **minimizes the variance** (risk) of the final hedged position.

C. Calculating the Minimum Variance Hedge Ratio (h^*)

The optimal hedge ratio is determined by the historical relationship between the change in the spot price of the exposed asset and the change in the futures price of the hedging asset.

Variable	Definition
ΔS	Change in the Spot Price of the asset being hedged (e.g., Jet Fuel) over the life of the hedge.
ΔF	Change in the Futures Price of the contract used (e.g., Heating Oil) over the life of the hedge.
σ_S	Standard deviation of ΔS .
σ_F	Standard deviation of ΔF .
ρ	Coefficient of correlation between ΔS and ΔF .

Formula for Minimum Variance Hedge Ratio (h^*):

$$h^* = \rho \frac{\sigma_S}{\sigma_F} \quad \text{or} \quad h^* = \frac{\text{Cov}(\Delta S, \Delta F)}{\sigma_F^2}$$

- **Intuition:** h^* is the slope of the best-fit line when plotting ΔS against ΔF .
 - If $\rho = 1.0$ and $\sigma_S = \sigma_F$ (prices move perfectly, dollar-for-dollar), then $h^* = 1.0$.
 - If $\rho = 1.0$ and $\sigma_F = 2\sigma_S$ (futures price moves twice as much as the spot price), then $h^* = 0.5$.
- **Hedge Effectiveness:** The proportion of variance eliminated by hedging is given by R^2 from the regression, which equals ρ^2 .

D. Calculating the Optimal Number of Contracts (N^*)

Once h^* is known, the optimal number of contracts (N^*) needed for the hedge is calculated as:

$$N^* = \frac{h^* \times Q_A}{Q_F}$$

Variable	Definition
N^*	Optimal number of futures contracts.
h^*	Minimum variance hedge ratio.
Q_A	Size of the position being hedged (units, e.g., gallons of jet fuel).
Q_F	Size of one futures contract (units, e.g., gallons of heating oil).

Example 3.3: Hedging Jet Fuel with Heating Oil

- **Exposure (Q_A):** Airline needs to buy **2 million gallons** of jet fuel.
- **Hedging Contract (Q_F):** Heating oil futures contract size is **42,000 gallons**.
- **Historical Data Analysis:**
 - σ_F (Std Dev of Futures ΔF): 0.0313
 - σ_S (Std Dev of Spot ΔS): 0.0263
 - ρ (Correlation ρ): 0.928

1. **Calculate h^* :**

$$h^* = 0.928 \times \frac{0.0263}{0.0313} \approx \mathbf{0.78}$$

2. **Calculate N^* :**

$$N^* = \frac{0.78 \times 2,000,000 \text{ gallons}}{42,000 \text{ gallons/contract}} \approx 37.14$$

3. **Result:** The airline should short **37 contracts** (rounded to the nearest whole number) to minimize the variance of its final purchase cost.

E. Tailing the Hedge (Adjusting for Daily Settlement)

The analysis for h^* above works perfectly for **forward contracts**. However, because **futures contracts** are marked-to-market daily, analysts sometimes refine the calculation using a concept called **tailing the hedge**.

- **Why Tailing?** Futures involve a series of one-day hedges where margin is settled daily. The change in the value of the spot position is being hedged by the change in the margin account balance.
- **The Tailing Formula (Value-Based):**

$$N^* = \hat{h} \frac{V_A}{V_F}$$

Variable	Definition
N^*	Optimal number of contracts.
$\hat{h}$	The refined hedge ratio based on <i>percentage</i> daily changes ($\hat{\rho}_{\frac{\Delta S}{\Delta F}}$).
V_A	Dollar Value of the Position Being Hedged ($S \times Q_A$).
V_F	Dollar Value of One Futures Contract ($F \times Q_F$).

- **Practicality:** While theory suggests changing the position daily to reflect current values of V_A and V_F , day-to-day changes in the number of contracts (N) are usually small and often ignored in practice.